Lecture Notes

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

In an eigenbasis an operator only multiplies each basis vector by its eigenvalue. So whatever we do with numbers (raise to a power, apply a function) we can do with a diagonalizable operator, separately for each eigenvalue. In this lecture we state exactly when an operator is *diagonalizable* and when the matrices of two operators are diagonal in one basis, and we prove the *Cayley–Hamilton theorem*. Then we compute *functions of operators*, above all the exponential e^{tA}, which solves the equations of linear dynamics.

Learning objectives

By the end of this lecture you will be able to:

- Decide whether an operator is diagonalizable, and find $A = PDP^{-1}$.
- Find a common eigenbasis of commuting diagonalizable operators.
- Apply the Cayley–Hamilton theorem to inverse matrices and high powers.
- Compute functions of an operator and their eigenvalues, especially e^{tA}.
- Solve $\dot{x} = Ax$ by expanding the initial state in an eigenbasis.

1 Diagonalizable operators

Definition 6.1 (Diagonalizable operator). An operator $T \in \mathcal{L}(V)$ is called *diagonalizable* if V has a basis consisting of eigenvectors of T. Equivalently, if the matrix $\mathcal{M}(T)$ is diagonal in some basis.

The two formulations agree, because the matrix of the operator T in a basis $v_1, \dots, v_n$ is diagonal exactly when $Tv_k = \lambda_k v_k$ for each k, i.e. when every basis vector is an eigenvector. The next theorem collects convenient conditions for diagonalizability.

Theorem 6.2 (Conditions for diagonalizability). Let $\dim V = n$, and let $\lambda_1, \dots, \lambda_m$ be all the distinct eigenvalues of an operator $T \in \mathcal{L}(V)$. Then the following statements are equivalent:

1. T is diagonalizable.
2. V has a basis of eigenvectors of T.
3. $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$.
4. $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

Proof. The first two statements are equivalent by definition. By the corollary "The sum of eigenspaces is direct" of Lecture 5, the sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is direct, so its dimension equals $\sum_k \dim E(\lambda_k, T)$ and is at most n.

Suppose V has a basis of eigenvectors. Each of its vectors lies in some eigenspace, so $\sum_k \dim E(\lambda_k, T) \geq n$. Together with the reverse inequality this gives equality of the dimensions.

If the dimensions add up to n, then the dimension of the direct sum equals $\dim V$, and a subspace of full dimension is the whole space V.

Finally, if V is the direct sum of the eigenspaces, then joining bases of the eigenspaces gives a basis of V consisting of eigenvectors. ∎

Corollary 6.3 (Distinct eigenvalues suffice). If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then T is diagonalizable.

Proof. Eigenvectors corresponding to distinct eigenvalues are linearly independent (Lecture 5). We have $\dim V$ linearly independent vectors, so they form a basis. ∎

This condition is sufficient but not necessary: the identity operator has the single eigenvalue 1, yet its matrix is already diagonal. For a *defective* operator the dimensions of the eigenspaces add up to less than $\dim V$, because some eigenspace is too small.

Example 6.4 (A diagonalizable operator). The matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

has distinct eigenvalues 3 and 1 (Lecture 5), so by Corollary 6.3 it is diagonalizable. Its eigenpairs are

$$\lambda_1 = 3,\ v_1 = (1,1) \quad \text{and} \quad \lambda_2 = 1,\ v_2 = (1,-1).$$

The matrix

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

has these eigenvectors as its columns. The matrix P is the change-of-basis matrix from the eigenbasis to the standard basis, so $P^{-1}AP$ is the matrix of the operator in the eigenbasis. The inverse formula for 2×2 matrices gives

$$P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Therefore

$$P^{-1}AP = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \operatorname{diag}(3,1).$$

The diagonal holds the eigenvalues, in the same order as the eigenvectors in the columns of P. ◁

Example 6.5 (A non-diagonalizable operator). The shear $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has the single eigenvalue 1, and its eigenspace $E(1,T) = \operatorname{span}\big((1,0)\big)$ is one-dimensional. The dimensions of the eigenspaces add up to $1 < 2$, so the shear is not diagonalizable: there is no basis of eigenvectors. ◁

Example 6.6 (A repeated eigenvalue that still diagonalizes). The matrix

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

is upper triangular. Therefore its eigenvalues are the diagonal entries $3, 2, 2$. The eigenspace $E(2,A) = \ker(A - 2I)$ is the solution set of the homogeneous system $(A - 2I)x = 0$. We reduce the matrix $A - 2I$:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies x_1 = -x_2.$$

So the free variables are x_2 and x_3. Setting $x_2 = 1,\ x_3 = 0$ reads off the vector $(-1, 1, 0)$; setting $x_2 = 0,\ x_3 = 1$ reads off the vector $(0, 0, 1)$. In the free-variable positions they carry $(1, 0)$ and $(0, 1)$, so neither is a multiple of the other. So the eigenspace

$$E(2, A) = \operatorname{span}\big((-1, 1, 0), (0, 0, 1)\big)$$

is *two-dimensional*. For $\lambda = 3$ we reduce the matrix $A - 3I$:

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 0 & \textcircled{1} & 0 \\ 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 \end{bmatrix} \Longrightarrow \quad x_2 = 0, \quad x_3 = 0.$$

Setting the free variable $x_1 = 1$, we get $E(3, A) = \operatorname{span}\big((1, 0, 0)\big)$. The eigenspace dimensions add up to $1 + 2 = 3$, so the matrix A is diagonalizable, although its eigenvalue 2 is repeated. Writing the eigenvectors into the columns of P, we get a diagonal matrix:

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \operatorname{diag}(3, 2, 2).$$

Compare the shear of Example 6.5: there the repeated eigenvalue had only a one-dimensional eigenspace. ◁

Diagonalization in practice. All the examples above follow one recipe, and it works for every diagonalizable operator: find the characteristic polynomial and its roots, find a basis of each eigenspace by Gaussian elimination, and write the basis vectors into the columns of a matrix P. In general, let T be a diagonalizable operator on $\mathbb{F}^n$, with eigenvectors $v_1, \dots, v_n$ and eigenvalues $\lambda_1, \dots, \lambda_n$. Let $A = \mathcal{M}(T)$ be its matrix in the standard basis. Form the matrices

$$P = \begin{pmatrix} | & \cdots & | \\ v_1 & \cdots & v_n \\ | & \cdots & | \end{pmatrix} \quad \text{(eigenvectors as columns)}, \qquad D = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

The equalities $Av_k = \lambda_k v_k$, written column by column, form the equality $AP = PD$. The matrix P is invertible, so

$$A = PDP^{-1}. \tag{6.1}$$

This is the change-of-basis formula of Lecture 4, applied to the eigenbasis (Figure 1). It lets us compute powers, and later functions, on the diagonal alone:

$$A^k = (PDP^{-1})^k = PD^kP^{-1} = P\,\operatorname{diag}(\lambda_1^k, \dots, \lambda_n^k)\,P^{-1},$$

because the middle factors $P^{-1}P$ cancel.

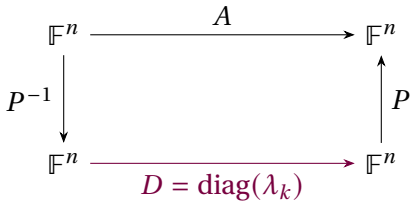


Figure 1: In the eigenbasis the operator is represented by the diagonal matrix D. The change of basis P gives $A = PDP^{-1}$. Powers and functions are applied to the diagonal entries of D.

Example 6.7 (Computing a power). In Example 6.4 we found the matrices P, P^{-1} and $D = \operatorname{diag}(3,1)$ for the matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, so

$$A^k = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 3^k & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 3^k+1 & 3^k-1 \\ 3^k-1 & 3^k+1 \end{pmatrix}.$$

Computing A^k directly for large k would take far longer. ◁

2 Commuting operators and simultaneous diagonalization

In physics we often need a basis in which the matrices of two operators are both diagonal. Such a basis can exist only if the operators commute. We begin with a single common eigenvector.

Proposition 6.8 (Commuting operators share an eigenvector). If V is a nonzero finite-dimensional complex vector space and $S, T \in \mathcal{L}(V)$ commute, then S and T have a common eigenvector.

Proof.

1. By the theorem "Eigenvalues exist over $\mathbb{C}$" of Lecture 5, the operator S has an eigenvalue λ.
2. Its eigenspace is invariant under T. Indeed, for $v \in E(\lambda, S)$,
$$S(Tv) = T(Sv) = T(\lambda v) = \lambda(Tv).$$
So $Tv \in E(\lambda, S)$.
3. The subspace $E(\lambda, S)$ is invariant, so we may regard T as an operator on this subspace. It is a nonzero complex space, so by the same theorem it contains an eigenvector v of T.
4. The vector v is an eigenvector of T. It lies in the subspace $E(\lambda, S)$, so it is also an eigenvector of S. ∎

Triangularization is not diagonalization. Repeating this argument with invariant subspaces, one can prove more (we do not prove it here):

$$\text{commuting complex operators} \implies \text{simultaneous upper triangularization.}$$

This means that there is a basis in which the matrices of both operators are upper triangular (Lecture 5, Section 6). Commuting is not enough for simultaneous diagonalization: both operators must also be diagonalizable. For example, the identity operator commutes with every operator, including the shear of Example 6.5. Yet these two operators cannot be diagonalized simultaneously.

Proposition 6.9 (Simultaneous diagonalization). Let $S, T \in \mathcal{L}(V)$ be diagonalizable. Their matrices are diagonal in one basis if and only if $ST = TS$. Such a basis consists of common eigenvectors of S and T.

Proof. If the matrices are diagonal in one basis, then $S = PD_1P^{-1}$ and $T = PD_2P^{-1}$ with the same matrix P, so

$$ST = PD_1D_2P^{-1} = PD_2D_1P^{-1} = TS,$$

because diagonal matrices commute. Conversely, let $ST = TS$. As in the proof of Proposition 6.8, S maps every eigenspace of T into itself.

1. Take any $v \in E(\lambda, S)$. The operator T is diagonalizable, so $v = w_1 + \cdots + w_r$ with $w_i \in E(\mu_i, T)$ for distinct μ_i (Theorem 6.2).
2. Then $0 = (S - \lambda I)v = (S - \lambda I)w_1 + \cdots + (S - \lambda I)w_r$, and the i-th term lies in $E(\mu_i, T)$. The sum of eigenspaces is direct (Lecture 5), so every term is zero: each nonzero w_i is a common eigenvector of S and T.
3. Every vector of V is a sum of eigenvectors of S, hence of common eigenvectors. By the theorem "Reduction to a basis" of Lecture 2, some of them form a basis of V, and in it both matrices are diagonal. ∎

In physics, commuting observables are called *compatible*: they have a common eigenbasis, so they can be measured precisely together.

Example 6.10 (Compatible and incompatible observables). Compare the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The eigenvectors of σ_z are $(1, 0)$ and $(0, 1)$, but neither of them is an eigenvector of σ_x. Multiplying the matrices, we get

$$\sigma_x \sigma_z = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \sigma_z \sigma_x.$$

Interpretation. By Proposition 6.9, commuting diagonalizable operators have a common eigenbasis. These two matrices do not commute and have no common eigenbasis. In physics this means that the spin projections on the x and z axes cannot be measured precisely together. This is the origin of the uncertainty relation, which we discuss in Lecture 9.

Each Pauli matrix does commute with the sum of their squares. With $\sigma_y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$ we have $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I$, so

$$\sigma_x^2 + \sigma_y^2 + \sigma_z^2 = 3I,$$

and $3I$ commutes with every matrix. Multiplied by $\hbar^2/4$, this sum is the square of the total spin, $\hat{S}^2 = \frac{3}{4}\hbar^2 I$. So we can know the square of the total spin and a single spin projection at the same time, for example the projection on the z axis. We cannot know two or more projections at the same time, because no two of the matrices σ_x, σ_y, σ_z have a common eigenvector: their eigenvectors are $(1, \pm 1)$, $(1, \pm i)$, and $(1, 0)$, $(0, 1)$. ◁

3 The Cayley–Hamilton theorem

The Cayley–Hamilton theorem relates an operator to its characteristic polynomial $\chi_T(\lambda) = \det(\lambda I - T)$. We prove it for diagonalizable operators and then apply it. The general proof rests on the upper-triangular matrix of Lecture 5; it closes the section, for self-study.

Theorem 6.11 (Cayley–Hamilton). Every operator T on a finite-dimensional vector space over $\mathbb{F}$ satisfies its own characteristic equation:

$$\chi_T(T) = 0.$$

Proof for diagonalizable T. Let A be the matrix of T in some basis. The operator T is diagonalizable, so, as in (6.1), $A = PDP^{-1}$, where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of T. For any polynomial p we have $p(A) = P\,p(D)\,P^{-1}$ and $p(D) = \operatorname{diag}(p(\lambda_1), \ldots, p(\lambda_n))$. Take $p = \chi_T$. Each eigenvalue λ_k is a root of the polynomial χ_T, so

$$\chi_T(A) = P \operatorname{diag}\big(\chi_T(\lambda_1), \ldots, \chi_T(\lambda_n)\big) P^{-1} = P \operatorname{diag}(0, \ldots, 0)\, P^{-1} = 0.$$

The matrix $\chi_T(A)$ is the matrix of the operator $\chi_T(T)$, so $\chi_T(T) = 0$. ∎

Corollary 6.12 (Inverse and powers as low-degree polynomials). Let T be invertible, and let $\chi_T(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$ (so $c_0 = (-1)^n \det T \neq 0$). Then

$$T^{-1} = -\frac{1}{c_0}\left(T^{n-1} + c_{n-1}T^{n-2} + \cdots + c_1 I\right).$$

More generally, every power T^k with $k \geq n$ can be written as a polynomial in T of degree less than n. Namely, $T^k = r(T)$, where r is the remainder of λ^k divided by $\chi_T(\lambda)$.

Proof. By the Cayley–Hamilton theorem, $T^n + \cdots + c_1 T + c_0 I = 0$. Moving $c_0 I$ to the other side and multiplying by $T^{-1}/(-c_0)$ gives the first claim. Divide λ^k by $\chi_T(\lambda)$ with remainder: $\lambda^k = q(\lambda)\chi_T(\lambda) + r(\lambda)$ with $\deg r < n$. Substituting T gives $T^k = q(T)\chi_T(T) + r(T) = r(T)$. ∎

Example 6.13 (Inverse matrix from Cayley–Hamilton). Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. Then $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$, so $A^2 - 4A + 3I = 0$ and

$$A^{-1} = \frac{1}{3}(4I - A) = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Check:

$$AA^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = I.$$

◁

Example 6.14 (Cayley–Hamilton, verified by hand). Let $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$. Then $\chi_A(\lambda) = \lambda^2 - 5\lambda - 2$, and indeed

$$A^2 - 5A - 2I = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The matrix A satisfies its own characteristic equation. ◁

Example 6.15 (Reducing a high power). In Example 6.14 we obtained $A^2 = 5A + 2I$. So we can write every higher power as a linear combination of the matrices A and I:

$$A^3 = A \cdot A^2 = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I,$$

and by induction $A^k = \alpha_k A + \beta_k I$ with scalars α_k, β_k. Thus any power of the matrix A is written with just two matrices, I and A. ◁

The general proof of Theorem 6.11 rests on the following lemma.

Lemma 6.16 (The product of the triangular factors is zero). Suppose T has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to a basis $v_1, \ldots, v_n$. Then

$$(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_n I) = 0.$$

Proof. The matrix is upper triangular, so each factor $T - \lambda_k I$ maps the vector v_k into a smaller subspace:

$$(T - \lambda_k I)v_k \in \operatorname{span}(v_1, \ldots, v_{k-1}).$$

We prove by induction on k that the product of the first k factors maps the vectors $v_1, \ldots, v_k$ to zero.

1. For $k = 1$, $(T - \lambda_1 I)v_1 = 0$.
2. Suppose the product of the first $k - 1$ factors maps the vectors $v_1, \ldots, v_{k-1}$ to zero.

3. The rightmost factor $T - \lambda_k I$ acts on the vector v_k first. It maps v_k into the subspace $\operatorname{span}(v_1, \dots, v_{k-1})$, which the remaining factors map to zero by the induction hypothesis. All the factors commute, because they are polynomials in the operator T. So we may apply the first $k-1$ factors first; they map the vectors $v_1, \dots, v_{k-1}$ to zero.
4. For $k = n$, the whole product maps every basis vector to zero, so it is the zero operator.

■

Proof of the Cayley–Hamilton theorem in general. Over $\mathbb{C}$ every operator has an upper-triangular matrix with diagonal entries $\lambda_1, \dots, \lambda_n$ (Lecture 5, where only a sketch of the proof is given). Then $\chi_T(\lambda) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n)$. By Lemma 6.16, $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$, and this product is exactly $\chi_T(T)$. Therefore $\chi_T(T) = 0$.

A real matrix can also be regarded as a complex matrix. For it the identity $\chi_T(T) = 0$ is already proved. Both sides of this identity are real matrices, so the identity also holds over $\mathbb{R}$. ■

4 Functions of operators

We have already used *polynomials* of an operator. To a diagonalizable operator T we can apply *any* function defined at its eigenvalues.

Definition 6.17 (Function of a diagonalizable operator). Let $T = PDP^{-1}$, where $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, and let the function f be defined at every point λ_k. Define

$$f(T) = P\, f(D)\, P^{-1}, \qquad f(D) = \operatorname{diag}\bigl(f(\lambda_1), \dots, f(\lambda_n)\bigr).$$

This definition agrees with the power-series definition whenever the series converges, and it does not depend on the choice of eigenbasis.

Both claims are easy to check. By Theorem 6.2, every vector splits uniquely as $v = \sum_\lambda v_\lambda$ with $v_\lambda \in E(\lambda, T)$. The matrix $f(D)$ multiplies each eigenbasis vector with eigenvalue λ by $f(\lambda)$, so $f(T)v = \sum_\lambda f(\lambda) v_\lambda$; no basis appears in this formula. For a power series $f(\lambda) = \sum_k a_k \lambda^k$, the equality $T^k v_\lambda = \lambda^k v_\lambda$ gives the same action term by term.

The most important example is the *exponential*. It is defined for every square matrix by the series

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

We read this series componentwise. Each of the n^2 entries is an ordinary complex power series whose terms are bounded by the terms of a scalar exponential series, so it converges. Nothing more is needed below. In Lecture 7 we introduce norms; then convergence can be studied for the matrices themselves rather than for their entries. For a diagonalizable matrix $A = PDP^{-1}$ the sum of the series is $e^A = P \operatorname{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})\, P^{-1}$.

Example 6.18 (Spin rotation: $e^{i\theta\sigma_x}$). Since $\sigma_x^2 = I$, the exponential series splits into even and odd powers:

$$e^{i\theta\sigma_x} = \sum_j \frac{(i\theta)^{2j}}{(2j)!} I + \sum_j \frac{(i\theta)^{2j+1}}{(2j+1)!} \sigma_x = \cos\theta\, I + i \sin\theta\, \sigma_x.$$

This operator rotates the spin about the x axis. The same calculation with any operator whose square is I gives the pair cos, sin. This is what the time evolution $e^{-i\hat{H}t/\hbar}$ of a two-level system looks like. ◁

How do the eigenvalues change when we apply a function to an operator?

Proposition 6.19 (Spectral mapping). If T is diagonalizable over $\mathbb{F}$ and $f : \operatorname{spec}(T) \to \mathbb{F}$, then

$$\operatorname{spec}\big(f(T)\big) = \{f(\lambda) : \lambda \in \operatorname{spec}(T)\},$$

and every eigenvector of T for λ is an eigenvector of $f(T)$ for $f(\lambda)$.

Proof. Let $Tv = \lambda v$. In eigen-coordinates the nonzero coordinates of v sit only in the positions where the diagonal of D holds λ, and $f(D)$ multiplies each of them by $f(\lambda)$. Therefore $f(T)v = Pf(D)P^{-1}v$ is the vector v multiplied by $f(\lambda)$. Conversely, the same eigenbasis also diagonalizes $f(T)$, and its diagonal holds exactly the values $f(\lambda_k)$. So $f(T)$ has no other eigenvalues. ∎

The converse fails: if f is not injective on $\operatorname{spec}(T)$, an eigenvector of $f(T)$ need not be an eigenvector of T. For example, for $T = \operatorname{diag}(1, -1)$ and $f(\lambda) = \lambda^2$ we get $f(T) = I$. Every nonzero vector is an eigenvector of I, while T has only two eigendirections.

Example 6.20 (Spectrum of a function). The spectrum of the matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ is $\{1, 3\}$. So the spectrum of e^A is $\{e,\ e^3\}$, and that of A^{10} is $\{1,\ 3^{10}\}$. The eigenvectors are the same throughout: $(1, -1)$ and $(1, 1)$. Once the eigenvectors are found, we need not recompute them for any function of A. ◁

5 Linear dynamics

In this section we put the lecture to work. The state of many physical systems changes in time by a linear law $\dot{x} = Ax$: the state is a vector $x(t)$, and its rate of change depends linearly on the state. Such equations describe a particle in a viscous liquid and quantum states, and the same idea solves the second-order equations of vibrations. We now have all the tools to solve them: the exponential gives the solution, diagonalization computes it, and the eigenvalues show at once how the system behaves. The key property of the exponential is the following.

Proposition 6.21 (Exponential and linear dynamics). For a fixed matrix A the function $t \mapsto e^{tA}$ satisfies

$$\frac{d}{dt} e^{tA} = A\, e^{tA} \quad \text{and} \quad e^{0A} = I.$$

Hence $x(t) = e^{tA}x_0$ is the unique solution of the equation $\dot{x} = Ax$ with the initial condition $x(0) = x_0$.

Proof. We differentiate the series term by term, componentwise:

$$\frac{d}{dt} \sum_k \frac{t^k A^k}{k!} = \sum_{k \geq 1} \frac{t^{k-1} A^k}{(k-1)!} = A\, e^{tA}.$$

This is allowed because each entry is a power series in t bounded by a scalar exponential; it is ordinary calculus applied n^2 times. Then $\frac{d}{dt}\big(e^{tA}x_0\big) = Ae^{tA}x_0$ and $x(0) = x_0$.

Uniqueness. Suppose $x(t)$ is any solution with the same initial value. The matrix A commutes with its power series, so

$$\frac{d}{dt}\big(e^{-tA}x(t)\big) = -Ae^{-tA}x(t) + e^{-tA}Ax(t) = 0.$$

Thus $e^{-tA}x(t) = x_0$. Likewise $\frac{d}{dt}\big(e^{tA}e^{-tA}\big) = 0$, because A commutes with e^{tA}, and at $t = 0$ this product equals I; hence $e^{tA}e^{-tA} = I$. Multiplying $e^{-tA}x(t) = x_0$ by e^{tA}, we get $x(t) = e^{tA}x_0$. ∎

Solution by modes. The whole matrix e^{tA} is rarely needed. Let $v_1, \dots, v_n$ be an eigenbasis of A with eigenvalues $\lambda_1, \dots, \lambda_n$. Expand the initial state in it: $x_0 = c_1 v_1 + \cdots + c_n v_n$. By spectral mapping (Proposition 6.19), $e^{tA} v_k = e^{\lambda_k t} v_k$, so

$$x(t) = e^{tA} x_0 = c_1 e^{\lambda_1 t} v_1 + \cdots + c_n e^{\lambda_n t} v_n. \tag{6.2}$$

Each term is a *mode*: its coefficient is multiplied by $e^{\lambda_k t}$. For a real eigenvalue the mode stays on its eigenline and its length changes by the factor $e^{\lambda_k t}$: a mode with $\lambda_k > 0$ grows, and one with $\lambda_k < 0$ decays. For a complex eigenvalue $\lambda_k = a + ib$ the factor $e^{\lambda_k t} = e^{at} e^{ibt}$ changes the size by e^{at} and turns the phase; we meet this case in Example 6.23. The coordinates c_k are found by solving the system $Pc = x_0$, so the inverse matrix P^{-1} is not needed.

In physics, passing to the eigenbasis *decouples* the equations. In the eigen-coordinates $y = P^{-1}x$ the coupled system $\dot{x} = Ax$ becomes a set of independent equations $\dot{y}_k = \lambda_k y_k$, with the solutions $y_k(t) = e^{\lambda_k t} y_k(0)$; this is formula (6.2) again.

Example 6.22 (Exponential of a 2×2 matrix and an initial value problem). For the matrix $A = PDP^{-1}$ of Example 6.7, where $D = \operatorname{diag}(3, 1)$,

$$e^{tA} = P \begin{pmatrix} e^{3t} & 0 \\ 0 & e^{t} \end{pmatrix} P^{-1} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^{t} & e^{3t} - e^{t} \\ e^{3t} - e^{t} & e^{3t} + e^{t} \end{pmatrix}.$$

Let us solve the equation $\dot{x} = Ax$ with the initial condition $x(0) = (1, 0)$. We expand the initial value in the eigenbasis:

$$x(0) = \tfrac{1}{2}(1, 1) + \tfrac{1}{2}(1, -1).$$

By (6.2), we multiply each mode by its exponential and add:

$$x(t) = \tfrac{1}{2} e^{3t}(1, 1) + \tfrac{1}{2} e^{t}(1, -1) = \tfrac{1}{2}\big(e^{3t} + e^{t},\ e^{3t} - e^{t}\big).$$

This is the first column of the matrix e^{tA}, as it must be, because $x(t) = e^{tA} x(0)$ by Proposition 6.21. After a long time the mode $e^{3t}(1, 1)$ dominates. ◁

How to read phase space. What does such a solution look like? Consider a particle moving in a plane through a very viscous liquid. The friction is so strong that we can neglect the particle's inertia: its velocity is proportional to the force acting on it. Let the matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ describe the force on the particle: at the point x the force is Ax (in suitable units). Then the particle's equation of motion is $\dot{x} = Ax$. This force pushes the particle away from the origin. The eigenpairs of A are $3,\ (1, 1)$ and $1,\ (1, -1)$, so on the eigenlines the force points straight away from the origin: on the line $\operatorname{span}\big((1, 1)\big)$ it equals $3x$, and on the line $\operatorname{span}\big((1, -1)\big)$ only x.

The state of the particle at time t is a point in the plane, $x(t) = \big(x_1(t), x_2(t)\big)$, and the equation $\dot{x} = Ax$ says in which direction and how fast this point moves. The set of all states is called the *phase space*. Here it is the plane, because in a viscous liquid the state of the particle is given by its position alone. Suppose that at time $t = 0$ the particle is at the point $(-2, 1)$, i.e. $x(0) = (-2, 1)$. Expand the initial state in the eigenbasis:

$$(-2, 1) = -\tfrac{1}{2}(1, 1) - \tfrac{3}{2}(1, -1).$$

By (6.2),

$$x(t) = -\tfrac{1}{2} e^{3t}(1, 1) - \tfrac{3}{2} e^{t}(1, -1).$$

What does this mean for the particle? Both terms grow, so the particle moves away from the origin. At first it moves straight to the left, because its velocity is $\dot{x}(0) = Ax(0) = (-3, 0)$. Then the particle turns downward, and at time $t = \frac{1}{2} \ln 3$ it crosses the x_1 axis at the point

$(-3\sqrt{3}, 0)$. The term with e^{3t} grows faster, so after a long time the particle moves almost parallel to the vector $(1, 1)$, to the left and down, i.e. in the direction in which the force is strongest. From then on its distance and its speed grow roughly like e^{3t}.

A drawing of the trajectories of several initial states is called a *phase portrait*. Every trajectory that does not start on an eigenline behaves in the same way (Figure 2): near the origin it is tangent to the slower eigendirection $(1, -1)$, and far from the origin it moves almost parallel to the faster direction $(1, 1)$.

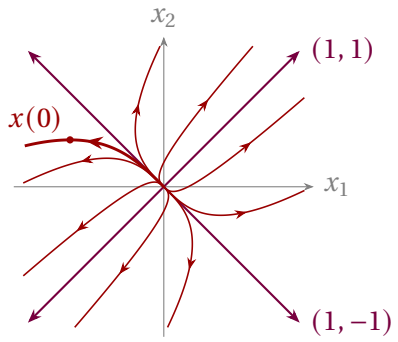


Figure 2: Phase portrait of the system $\dot{x} = Ax$ for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. Each curve is the trajectory of one initial state, and the arrow shows the direction of motion. The highlighted trajectory passes through the point $x(0) = (-2, 1)$.

So far the eigenvalues were real, and the modes grew along the eigenlines. What happens when the eigenvalues are complex?

Example 6.23 (Imaginary eigenvalues produce oscillation). Let $\omega > 0$. The eigenvalues of the matrix $A = \left(\begin{smallmatrix} 0 & -\omega \\ \omega & 0 \end{smallmatrix}\right)$ are $\pm i\omega$. Since $A^2 = -\omega^2 I$, the exponential series splits into even and odd powers, as in Example 6.18, and

$$e^{tA} = \begin{pmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{pmatrix}.$$

So the equation $\dot{x} = Ax$ describes rotation at angular frequency ω. For example, when $x(0) = (1, 0)$, the state $x(t) = (\cos \omega t, \sin \omega t)$ goes round a circle, and both of its coordinates oscillate harmonically (Figure 3). Imaginary eigenvalues produce oscillation, while the real ones in Example 6.22 produced growth. This is how linear algebra describes harmonic oscillation. ◁

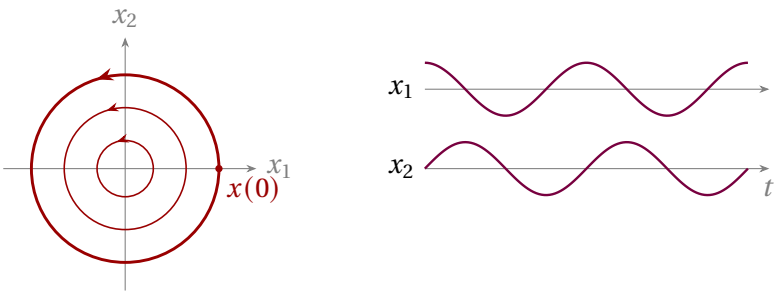


Figure 3: Phase portrait of the system $\dot{x} = Ax$ for $A = \left(\begin{smallmatrix} 0 & -\omega \\ \omega & 0 \end{smallmatrix}\right)$: every trajectory other than the origin, which stays put, is a circle traversed counterclockwise. The highlighted trajectory starts at the point $x(0) = (1, 0)$; its coordinates $x_1(t) = \cos \omega t$ and $x_2(t) = \sin \omega t$ oscillate harmonically.

The same idea works for vibrations. The state of a vibrating system includes the velocities as well as the displacements, and its equation $\ddot{x} = -Kx$ is of the second order. Yet it decouples in the same way: in the eigen-coordinates $y = P^{-1}x$ of K it becomes $\ddot{y}_k = -\kappa_k y_k$, where the κ_k are the eigenvalues of K. For $\kappa_k > 0$ these are independent oscillations, the *normal modes*, with angular frequencies $\sqrt{\kappa_k}$.

Example 6.24 (Normal modes of two coupled masses). Two equal masses are joined to each other by a spring, and each is tied to a wall by an identical spring. Their displacements obey the equation $\ddot{x} = -Kx$ with $K = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ (in suitable units). The eigenvalues of K are 1 and 3, and the eigenvectors are $(1,1)$ and $(1,-1)$. In these *normal coordinates* the system splits into two independent oscillators. The masses move together (the mode $(1,1)$) at angular frequency $\omega = 1$, or in opposite directions (the mode $(1,-1)$) at frequency $\omega = \sqrt{3}$ (Figure 4). Every motion is a sum of these modes. For example, pull the first mass aside by 1, hold the second in place, and release both from rest. Then $x(0) = (1,0) = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$ and

$$x(t) = \tfrac{1}{2}\cos t\,(1,1) + \tfrac{1}{2}\cos\sqrt{3}\,t\,(1,-1).$$

The oscillation passes back and forth between the two masses. So diagonalizing the matrix K is exactly finding the normal modes. ◁

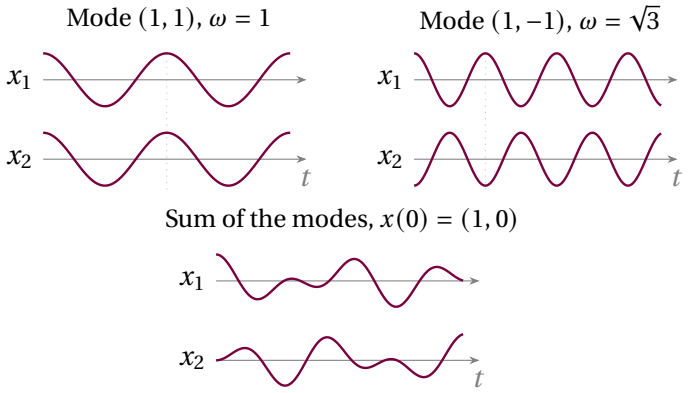


Figure 4: The normal modes of two coupled masses: the displacements $x_1(t)$ and $x_2(t)$ of the masses. In the mode $(1,1)$ the masses oscillate together, $x_1 = x_2 = \cos t$. In the mode $(1,-1)$ they oscillate in opposite directions, $x_1 = -x_2 = \cos\sqrt{3}\,t$. The dotted line marks the moment when x_1 is back at its maximum. The sum of the modes for $x(0) = (1,0)$ and $\dot{x}(0) = 0$: the oscillation passes from the first mass to the second and back.

Remark 6.25 (Quantum time evolution). A quantum state obeys the Schrödinger equation $i\hbar\frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle$. If the Hamiltonian $\hat{H}$ does not depend on time, this is the equation $\dot{x} = Ax$ with the fixed matrix $A = -i\hat{H}/\hbar$, so by Proposition 6.21 the state evolves by the law

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle.$$

In the energy eigenbasis $\hat{H} = \operatorname{diag}(E_1,\dots,E_n)$, so

$$e^{-i\hat{H}t/\hbar} = \operatorname{diag}\big(e^{-iE_1t/\hbar},\dots,e^{-iE_nt/\hbar}\big).$$

Over time, each state of definite energy only acquires a phase factor $e^{-iE_nt/\hbar}$, and a general state is a superposition of such states. So diagonalizing the Hamiltonian means solving the dynamics. That the Hamiltonian can always be diagonalized we prove in Lecture 10 (the spectral theorem).

When diagonalization fails *[self-study]*

Not every operator is diagonalizable. In that case *generalized eigenvectors*, which we define below, come to the rescue. The obstruction to diagonalization is the *nilpotent* part of the operator. An operator N is called nilpotent if $N^k = 0$ for some k; no change of basis can remove such a part. Suppose an operator has a single eigenvalue λ and $A = \lambda I + N$, where N is nilpotent. The matrices λI and N commute, so the exponential factors, and the series for e^{tN} *terminates*:

$$e^{tA} = e^{\lambda t}\, e^{tN} = e^{\lambda t}\Big(I + tN + \tfrac{t^2}{2!}N^2 + \cdots + \tfrac{t^{k-1}}{(k-1)!}N^{k-1}\Big).$$

Here we used $e^{t(B+C)} = e^{tB}e^{tC}$ for commuting matrices B and C. Indeed, $F(t) = e^{tB}e^{tC}$ satisfies $F' = BF + e^{tB}Ce^{tC} = (B+C)F$, because C commutes with e^{tB}, and $F(0) = I$; by the uniqueness in Proposition 6.21, applied to each column, $F(t) = e^{t(B+C)}$. For the shear-type matrix $A = \left(\begin{smallmatrix}\lambda & 1\\ 0 & \lambda\end{smallmatrix}\right)$ the nilpotent part $N = \left(\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right)$ satisfies $N^2 = 0$, so

$$e^{tA} = e^{\lambda t}\begin{pmatrix}1 & t\\ 0 & 1\end{pmatrix}.$$

The extra factor t shows that the eigenvalue is defective. It gives the solutions $t\,e^{\lambda t}$, the same ones that appear when the characteristic equation of a linear differential equation has a repeated root.

Generalized eigenvectors.

Definition 6.26 (Generalized eigenvectors and eigenspaces). A nonzero vector v is called a *generalized eigenvector* for the eigenvalue λ if

$$(T - \lambda I)^k v = 0 \qquad \text{for some } k \geq 1.$$

The *generalized eigenspace* is the set

$$G(\lambda, T) = \{v : (T - \lambda I)^k v = 0 \text{ for some } k\}.$$

Clearly,

$$E(\lambda, T) \subseteq G(\lambda, T).$$

There is no need to search for the exponent k: one fixed exponent is enough. Write $N = T - \lambda I$ and $n = \dim V$. The kernels grow: $\ker N \subseteq \ker N^2 \subseteq \cdots$. This chain stops as soon as two consecutive kernels coincide. Indeed, suppose $\ker N^j = \ker N^{j+1}$ and $N^{j+2}v = 0$. Then $Nv \in \ker N^{j+1} = \ker N^j$, so $N^{j+1}v = N^j(Nv) = 0$. Hence $\ker N^{j+1} = \ker N^{j+2}$, and the same step repeats from there on. Until the chain stops, every inclusion is strict, so at each step the dimension grows by at least 1. The kernels are subspaces of V, so the chain stabilizes by the n-th step at the latest. Therefore

$$G(\lambda, T) = \ker(T - \lambda I)^{\dim V}.$$

Theorem 6.27 (Generalized eigenspace decomposition · cited). Let V be a finite-dimensional complex vector space, and let $\lambda_1, \ldots, \lambda_r$ be all the distinct eigenvalues of an operator T. Then for each eigenvalue λ the dimension $\dim G(\lambda, T)$ of its generalized eigenspace equals its algebraic multiplicity, and V is the direct sum of the generalized

eigenspaces:

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_r, T).$$

Interpretation. A defective operator may lack ordinary eigenvectors, but there are always enough generalized eigenvectors to span the whole space. This decomposition is the foundation of the Jordan form. We do not prove the theorem.

Example 6.28 (A generalized eigenvector). The only eigenspace of the shear $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is

$$E(1, T) = \operatorname{span}((1, 0)).$$

However,

$$(T - I)^2 = 0, \qquad G(1, T) = \ker(T - I)^2 = \mathbb{C}^2.$$

For example, the operator $T - I$ maps the vector $(0, 1)$ to the eigenvector $(1, 0)$, and this one to the zero vector:

$$(0, 1) \xmapsto{T-I} (1, 0) \xmapsto{T-I} 0.$$

Such a sequence of vectors is called a *Jordan chain*. It leads from a generalized eigenvector to an ordinary eigenvector. ◁

Theorem 6.29 (Jordan normal form · cited). Every operator on a finite-dimensional complex space has a basis of generalized eigenvectors in which its matrix is block diagonal, each block being a *Jordan block*

$$J = \lambda I + N, \qquad N = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix}.$$

The block has the eigenvalue λ on its diagonal, and above the diagonal the ones of the nilpotent shift N. Each block corresponds to one Jordan chain of generalized eigenvectors (Example 6.28).

The shift N maps each basis vector of the block to the previous one, $e_j \mapsto e_{j-1} \mapsto \cdots \mapsto e_1 \mapsto 0$. After finitely many steps only the zero vector remains; this is what it means for N to be nilpotent.

Theorem 6.30 (Jordan–Chevalley decomposition · cited). On a finite-dimensional complex vector space every operator A can be written uniquely as a sum

$$A = D + N, \qquad DN = ND,$$

where D is diagonalizable, N is nilpotent, and D and N are polynomials in A.

Since D and N are polynomials in A, they commute with every matrix that commutes with A. So the decomposition preserves all the symmetries of A. It is also the most convenient way to compute $e^{tA} = e^{tD}e^{tN}$ for any matrix: the diagonalizable part gives oscillation and decay, and the nilpotent part gives corrections polynomial in t.

Remark 6.31 (Nilpotent operators and ladder operators). In physics, nilpotent operators appear as ladder operators on finite-dimensional spaces. On the state space of spin j (of dimension $2j + 1$) the raising operator J_+ maps each lower state $|m\rangle$ to a nonzero multiple

of $|m+1\rangle$, and it maps the top state to zero. Therefore $J_+^{2j+1} = 0$: J_+ is nilpotent, and after rescaling the basis vectors its matrix is the shift N of a single Jordan block. The lowering operator of the harmonic oscillator, truncated to N levels, is nilpotent in the same way ($a^N = 0$). The untruncated oscillator is discussed in Lectures 13–14.

Remark 6.32 (Where defective operators occur in physics). Operators corresponding to observables are always diagonalizable (Lectures 9–10), so there are no defective operators among them. Defective operators occur in systems that lose energy. For example, the repeated root of a critically damped oscillator gives the solution $t\,e^{-\gamma t}$, similar to the $t\,e^{\lambda t}$ obtained above. Here we only state the Jordan form and the Jordan–Chevalley decomposition, without proving their construction. Later we use only the principle: diagonalize when possible, and otherwise add a nilpotent correction.

Summary of main concepts

An operator is diagonalizable if and only if it has an eigenbasis, i.e. if the dimensions of its eigenspaces add up to $\dim V$. A convenient sufficient condition: $\dim V$ distinct eigenvalues. Commuting diagonalizable operators are diagonal in one common basis.

Every operator satisfies its own characteristic equation $\chi_T(T) = 0$ (the Cayley–Hamilton theorem). With it we write the inverse matrix and high powers as low-degree polynomials.

In an eigenbasis $A = PDP^{-1}$, so powers and functions are computed separately for each eigenvalue: $A^k = PD^kP^{-1}$ and $f(T) = Pf(D)P^{-1}$. Spectral mapping sends an eigenvalue λ to $f(\lambda)$.

The exponential e^{tA} solves the equation $\dot{x} = Ax$, and in an eigenbasis the solution is a sum of modes $c_k e^{\lambda_k t} v_k$. So in quantum mechanics, once the Hamiltonian is diagonalized, we immediately have the time evolution $e^{-i\hat{H}t/\hbar}$.

- **Diagonalizable** — has an eigenbasis $\iff$ the eigenspace dimensions add up to $\dim V$; $\dim V$ distinct eigenvalues suffice.
- **Diagonalization in practice** — $A = PDP^{-1}$, so powers and functions are computed separately for each eigenvalue.
- **Commuting operators** — on a complex space they share an eigenvector; commuting diagonalizable operators are diagonal in one basis.
- **Cayley–Hamilton theorem** — $\chi_T(T) = 0$; the inverse matrix and high powers are written as low-degree polynomials in T.
- **Spectral mapping** — f sends an eigenvalue λ to $f(\lambda)$, and the eigenvectors stay the same.
- **Linear dynamics** — $x(t) = e^{tA}x_0$ solves $\dot{x} = Ax$; in an eigenbasis it is a sum of modes $c_k e^{\lambda_k t} v_k$; diagonalizing H gives $e^{-iHt/\hbar}$.
- **Jordan normal form** — this form exists even when diagonalization fails (*self-study*).